Housing First or Treatment First?

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Response to Referee 2

Conceptual and detail concerns, responses in red

Thank you for the report, and in particular for the two welfare margins in your second paragraph. One of them has become a feature of the revised model and the other has become an intergenerational linkage that the Associate Editor independently asked for. Both changes came from this report.

Conceptual concerns

C1 The welfare results rest on the elasticity of drug use to housing access

> *"... all of the welfare results rest on whether the model has an accurate elasticity of drug use to housing access. The only empirical drug use elasticity mentioned is a price elasticity of drug use participation, but it isn't clear to me that this provides enough or the correct discipline for these policy experiments. I would like to see that the model matches some evidence on the response of drug use to housing access (see Milby, et al (2005) for cocaine users, but there are many papers on this topic.)"*

This is the crux, Referee 3 and the Associate Editor agree, and it is now a calibration target rather than an untested implication.

Milby et al. (2005) is the right trial for it, and it turns out to identify two things rather than one, which resolved a separate problem in the model.

- The *contrast between arms* identifies the response of abstinence to housing access, which is your requested elasticity.
- The *level in the no housing arm*, 4.48 consecutive weeks abstinent out of 24, or 18.7 per cent among homeless adults with a substance use disorder, identifies how costly abstention is for a household that already has the disorder. That parameter was previously asserted to be small enough to keep the drug choice interior, which is a statement about numerical convenience rather than about data. It also left the housing access moment undefined: if abstention is impossible for a household with a disorder, the no housing arm abstains zero per cent of the time and the ratio between arms is zero over zero.

One tension in this literature is worth reporting, because it looks like a contradiction and is not. Milby finds housing raises abstinence. The At Home / Chez Soi trial finds unconditional housing does nothing to substance use problem severity, 1.04 with a confidence interval of 0.89 to 1.21. Milby's housing was *contingent on abstinence*; At Home / Chez Soi's was *unconditional*. Contingency is doing the work, not housing. That distinction is exactly the paper's subject, and the two trials together are better evidence for it than either alone. This is now in the text.

[Built since the first draft of this letter.] It is now in the code and not only in the written model. The Stone-Geary shifter depends on the housing chosen, and every place in the program that evaluates the drug term goes through a single function, so the flow utility, its marginal utility

and the two closed form splits of consumption against drugs cannot disagree with one another. That was the implementation risk worth spending care on: had those been patched separately, the objective and its own first order condition would have drifted apart silently. The parameter is bracketed on the contrast between Milby's arms, as described above.

C2 A large homeless population is a negative externality

> *"... a large homeless population, especially if many are addicted to drugs, creates a negative externality beyond the fiscal burden of providing services. ... One structural rental housing paper that considers the negative externality of homelessness (though doesn't have drug addiction) is Corbae, et al (2024). ... the externality seems very easy to introduce into the current model (though would require a new calibration moment)."*

Adopted, and it does more work than you may have anticipated, because it lets us delete something that three of the four reports objected to.

A disutility from the size of the homeless population now enters period utility. This replaces the loss of consortium adjustment entirely. Referee 1's comments 3(a) and 3(b) said that adjustment was applied after the model was solved, was therefore internally inconsistent, and rested on a parameter whose conservative value is zero. The Associate Editor's comment A9 said the same. Your externality answers all of those objections at once, because it sits *inside* the value function where households internalise it.

It also gives Referee 3's comment M2c a concrete answer. He asks where the market failure comes from if drug use is rational and forward looking. There are now two answers in the model: the fiscal externality, which was always there, and this one, which is not fiscal and which no individual household internalises when it chooses whether to use.

On Corbae, Glover and Nattinger: we cite them, and we are explicit about what does and does not transfer. Their externality is about neighbourhood housing *quality* and is identified off rent decontrol, so their functional form is not the right one to import. What transfers is the identification logic. We use a contingent valuation approach for the calibration moment instead.

[Built since the first draft of this letter.] **[Conceded.]** We have changed our minds on this one, and against you, so it is worth being explicit rather than quiet. The disamenity is now *absent* from the model rather than present and set to zero in the benchmark.

The reason is the size of the object. Loubière et al. (2020) ask adults in eight European countries what they would pay in annual taxes to end homelessness through Housing First, and get €23 a year on average and €57 among the 42 per cent willing to pay anything, which is 0.08 and 0.20 per cent of average annual earnings. A term that small cannot change a ranking. Carrying it, however, puts the aggregate homelessness rate inside the utility function, which makes it an argument the distribution itself determines, and so adds a fixed point to the equilibrium in exchange for nothing.

What we have done instead is state the direction of the omission. Leaving the disamenity out can only *understate* the case for whichever policy houses more people, so every result we report on that margin is a lower bound. If you would still prefer it carried and reported across a range, it is a small change to restore and we will restore it.

C3 Parents suffer anxiety about their children

> *"... parents suffer anxiety that their children may become drug addicts and/or homeless in adulthood. ... the dynastic element seems like more of a stretch, but could be discussed (I think it would only*

amplify the welfare gains from reducing addiction).”

We have done more than discuss it, because the Associate Editor’s comment A6 pushes in the same direction from a different angle: he objects that the previous draft discussed intergenerational transmission in section 4.1.3 without any intergenerational linkage in the model, which made the discussion decorative. Your comment supplies exactly the linkage that would make it load bearing.

Newborn productivity is now drawn from a parent’s productivity through a persistent process, and the parent’s period utility carries a warm glow term in the expected newborn value of their child. This is the cheapest structure that makes the transmission real: it does not require tracking dynasties or adding a state, and it means the persistence parameter matters for welfare rather than appearing only in a sentence.

We report results with the warm glow weight set to zero as well, so a reader can see how much of the welfare result rests on it. Your prediction is that it amplifies rather than reverses, and that is also our prior, but it is now something the paper shows rather than something we assert. The weight itself is disciplined against the literature on altruism towards children rather than chosen.

C4 The rich model is a black box; show the policy functions

> *“. . . it can be a bit of a black box. It would be helpful to have a section where the mechanisms of the model are explored more thoroughly. For example, it would be useful to show and explain intuitively a plot of the drug use policy function for addicts and non-addicts as their income and wealth vary.”*

Done, as the figure you describe, and it is now part of a broader mechanism section that also serves Referee 1’s comment 1(d) and Referee 3’s comment M2b.

The section has three parts: the policy function figure you ask for; a channel shutdown ladder that re solves the model with one channel disabled at a time and reports what happens to the income gradient of use; and a decomposition of the sources of the externality. The three together are the paper’s answer to the charge that the model is opaque, and they replace the static appendix that Referee 1 found was claiming more than it established.

C5 The utility specification bakes in an income contingent gain or loss

> *“It would also help to further explain the importance of the period utility specification in (13) which, I think, implies that drug addiction is more attractive for low income people . . . it also bakes in an income-contingent gain/loss from becoming addicted by shifting the utility weight towards drug consumption away from non-drug consumption and housing.”*

[Conceded.] You and Referee 3, in his comment M2a, identify the same problem, and it is a serious one: the previous specification made reducing substance use disorder and expanding housing mechanical welfare complements even with no incentive effect at all. Since the complementarity between generosity and conditionality is the paper’s headline result, the specification was doing the work that the economics was supposed to do.

Preferences are now *fixed* across substance use disorder status: one set of weights, one subsistence parameter. Nothing about the utility function changes when a household acquires the disorder.

The Associate Editor suggested a milder version, keeping the weights invariant but letting the subsistence parameter vary by status. We tried that and it does not reproduce the target the

specification was carrying, namely the gap between 44 per cent of income spent on drugs among people with a disorder and 11 per cent among experimenters: the subsistence parameter shifts the level of drug spending, but the ratio is governed by the curvature of the subsistence term, and opening a fourfold gap by moving it drives experimenters to a corner.

With fully fixed preferences that gap has to emerge endogenously, and it does, from forward looking restraint. A household without a disorder limits its use because use raises the probability of acquiring one. A household that already has one faces no such margin. The spending gap becomes a prediction rather than an assumption, and the welfare complementarity becomes an incentive effect rather than an artefact of the utility function. This is a stronger result than the one it replaces and we are grateful for the push.

[Built since the first draft of this letter.] Done in the code, where it previously was not. The switch that raised the weight on drugs from 0.13 to 0.45 for a household with the disorder is off, so the weights on consumption, housing and drugs are now the same function clean and addicted, and the disorder acts on the drug decision only through the Stone-Geary shifter. The consistency check inside the program that used to demand the two shifters be equal has been relaxed accordingly, since the shifter is now the one thing that is meant to differ.

C6 Optimal voucher design, and χ_H looks high

> *"The parameter χ_H governs the fiscal cost of having somebody unhoused and appears pretty high at 25% of average income. Would there be significant further welfare gains from broadening the voucher program by making higher incomes eligible or covering more of the rent (with the drug-free contingency)? My (perhaps wrong) reading is that the cost of having somebody unhoused is significantly higher than the rent on the smallest housing unit."*

Your reading is correct, it is not wrong, and it is the mechanism behind the near self financing result. The paper should be transparent about that rather than let it emerge as a surprise, so it is now stated in the text where the result is presented, and the sensitivity is in the body rather than an appendix.

On the level of χ_H, the Associate Editor's comment A15 points to Abramson and Van Nieuwerburgh as a comparison. Having read their paper, there is a point of fact worth recording: their 2,500 dollars per household per month for the literally homeless, and half that for the doubled up, is an *assumption* inside their calibration, stated as such with components listed in their footnote 11, rather than an independent estimate. So it cannot validate our number. We report the comparison, because a reader will want it, described accurately as a comparison of two calibration choices. Blended across their two groups it comes to about 1,775 dollars per month.

On optimal design, we now run the continuous sweep that Referee 3's minor comment 3 also asks for: the probability of voucher receipt is varied from the benchmark to universal, and welfare, homelessness, use and the labour tax are plotted along it. This shows whether universal provision is optimal or merely a corner, which is his question, and it shows where the gains from broadening eligibility run out, which is yours. We run the sweep separately with and without the sobriety condition, so the cross partial that the paper's headline rests on is visible as the gap between two curves rather than as a comparison of four table cells.

Detail concerns

D1 The housing grid

> *"... the housing grid. It has seven values but only three are calibrated ... The full grid is then listed in footnote 4 rather than a calibration table. My assumption before seeing the grid was that the bounds were calibrated and then the sizes were distributed uniformly ... But they are not uniformly spread out, at least in levels or ratios. So, how are the housing levels being chosen and does it matter for results? Could they just be spread uniformly without affecting the calibration and policy conclusions? If not then I think they need to be included as parameters in the calibration."*

The honest answer to the original question is that the grid was not chosen by a principle we could defend, which is why the footnote existed. The revision removes the problem rather than justifying it.

With homeownership cut at the Associate Editor's request in his comment A4, the housing state is homeless plus three rental sizes. Every point of that grid is now pinned by a moment, and the moments are the size distribution of occupied units from the American Community Survey, tenure by bedrooms: 13.21, 24.12 and 62.67 per cent across zero to one, two, and three or more bedrooms. All households rather than renters only, because the model has no homeownership and housing is a pure flow, so an owner's analogue is imputed rent and the distribution over all occupied units is the right object. The middle category is the residual and is not targeted.

The grid now appears in the calibration table with its moments, not in a footnote. As a by product, the two housing cost burden moments that previously identified the grid are freed to serve as out of sample validation, which is a stronger thing to show than a moment the grid was fitted to.